

# JEE MAIN REVISION MASTER: PARABOLA SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 52 Analyzed PYQs

## 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                                   | Frequency | Category |
|---------------|------------------------------------------------|-----------|----------|
| Very High     | Standard Parabola, Focus, Directrix & Tangents | 51 PYQs   | Parabola |
| Standard      | Normals to Parabola & Co-Normal Points         | 2 PYQs    | Normals  |

## 2. Master Formula Reference Card

| Concept / Topic                                | Mathematical Formulation & Governing Equations                                                                                                              |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Standard Parabola, Focus, Directrix & Tangents | $y^2 = 4ax \implies \text{Focus} (a, 0), \text{Directrix: } x = -a, \text{LR} = 4a, \text{Tangent: } y = mx + \frac{a}{m}, \text{Parametric: } (at^2, 2at)$ |
| Normals to Parabola & Co-Normal Points         | $\text{Normal: } y = mx - 2am - am^3, \text{Parametric: } y + tx = 2at + at^3, \sum m_i = 0$                                                                |

## 3. Deep-Dive Concepts & Linked PYQ Evidence

### 3.1 Standard Parabola, Focus, Directrix & Tangents (Very High Yield — 51 PYQs)

**Core Principle:** Focal properties of  $y^2 = 4ax$ , slope form of tangent  $y = mx + a/m$ , and parametric chords.

■ **Caution / JEE Pitfall:** For  $x^2 = 4ay$ , the tangent in slope form is  $y = mx - am^2$ , NOT  $y = mx + a/m$ !

■ **Speed Trick:** The tangents at the extremities of any focal chord intersect at right angles ( $90^\circ$ ) strictly on the DIRECTRIX.

**Representative Source PYQs:**

- [2026 Evening Shift] Q#1 (MCQ): "Let  $O$  be the vertex of the parabola  $y^2=4x$  and its chords  $OP$  and  $OQ$  are perpendicular to each other. If the locus of the mid-point of  $th...$ " [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Let the parabola  $y=x^2+px+q$  passing through the point  $(1,-1)$  be such that the distance between its vertex and the  $x$ -axis is minimum..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Let  $O$  be the vertex of the parabola  $x^2=4y$  and  $Q$  be any point on it. Let the locus of the point  $P$ , which divides the line segment  $OQ$  in..." [Ans: N/A]

### 3.2 Normals to Parabola & Co-Normal Points (Standard Yield — 2 PYQs)

**Core Principle:** Cubic normal equation  $am^3 + (2a - x)m + y = 0$ , maximum of three normals from external point, and co-normal properties.

■ **Caution / JEE Pitfall:** From an external point  $(h, k)$ , three normals can be drawn to  $y^2 = 4ax$  if and only if  $h > 2a$ .

■ **Speed Trick:** The algebraic sum of the slopes of three co-normal points is always ZERO:  $m_1 + m_2 + m_3 = 0$ .

**Representative Source PYQs:**

- **[2024 Morning Shift] Q#1 (MCQ):** "If the shortest distance of the parabola  $y^2=4x$  from the centre of the circle  $x^2+y^2-4x-16y+64=0$  is  $d$ , then  $d^2$  is equal to..." **[Ans: N/A]**
- **[2023 Morning Shift] Q#3 (MCQ):** "The distance of the point  $(6, -2\sqrt{2})$  from the common tangent  $y=mx+c, m>0$ , of the curves  $x=2y^2$  and  $x=1+y^2$  is ..." **[Ans: N/A]**

## 4. Benchmark Illustrative PYQs with Master Solutions

### Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

Let  $O$  be the vertex of the parabola  $y^2=4x$  and its chords  $OP$  and  $OQ$  are perpendicular to each other. If the locus of the mid-point of the line segment  $PQ$  is a conic  $C$ , then the length of its latus rectum is :

- (A) 1
- (B) 2
- (C) 4
- (D) 8

**Correct Answer: None**

**Solution Analysis:**  $t_1 t_2 = -4$   $2 = t_1^2 + t_2^2$   $2k = 2(t_1 + t_2)$   $2 = k^2 - 2t_1 t_2$   $2 = k^2 + 8$   $k^2 = -6$   $k = \pm\sqrt{-6}$   $L.R. = 2$

### Benchmark Problem 2: JEE Main 2023 (Online) 25th January Morning Shift (Q#3)

The distance of the point  $(6, -2\sqrt{2})$  from the common tangent  $y=mx+c, m>0$ , of the curves  $x=2y^2$  and  $x=1+y^2$  is :

- (A)  $(1/3)$
- (B) 5
- (C)  $(14/3)$
- (D)  $5\sqrt{3}$

**Correct Answer: None**

**Solution Analysis:**  $y^2 = (x/2) \Rightarrow$  tangency  $= mx + (1/8m)$   $y^2 = x - 1 \Rightarrow$  tangency  $= m(x-1) + (1/4m)$  For common tangent  $(1/8m) = -m + (1/4m) \Rightarrow 1 = -8m^2 + 2$   $m > 0 \Rightarrow m = \frac{1}{2\sqrt{2}}$   $\Rightarrow$  Common tangent is  $y = \frac{x\sqrt{2}}{2} + \frac{1\sqrt{2}}{2} \Rightarrow x - 2\sqrt{2}y + 1 = 0$  Distance of point  $(6, -2\sqrt{2})$  from common tangent  $= 5$